



Java

```
class Car {  
    String color;  
    String model;  
  
    void displayDetails() {  
        System.out.println("Car Model: " +  
model + ", Color: " + color);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Car car1 = new Car();  
        car1.model = "Honda Civic";  
        car1.color = "Red";  
        car1.displayDetails();  
    }  
}
```



Java

```
class Student {
    String name;
    int age;

    // Constructor
    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }

    void showInfo() {
        System.out.println("Name: " + name + ",
Age: " + age);
    }
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student("Alice", 20);
        s1.showInfo();
    }
}
```



Java

```
class BankAccount {
    private double balance;

    public void setBalance(double balance) {
        if (balance >= 0) {
            this.balance = balance;
        } else {
            System.out.println("Invalid
balance");
        }
    }

    public double getBalance() {
        return balance;
    }
}

public class Main {
    public static void main(String[] args) {
        BankAccount account = new
BankAccount();
        account.setBalance(5000);
        System.out.println("Balance: " +
account.getBalance());
    }
}
```



Java

```
class Employee {
    private int age;

    public void setAge(int age) {
        if(age > 18) {
            this.age = age;
        } else {
            System.out.println("Age must be
above 18");
        }
    }

    public int getAge() {
        return age;
    }
}

public class Main {
    public static void main(String[] args) {
        Employee emp = new Employee();
        emp.setAge(25);
        System.out.println("Employee Age: " +
emp.getAge());
    }
}
```



Java

```
class Animal {  
    void eat() {  
        System.out.println("Animal eats food");  
    }  
}  
  
class Dog extends Animal {  
    void bark() {  
        System.out.println("Dog barks");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Dog dog = new Dog();  
        dog.eat();  
        dog.bark();  
    }  
}
```



Java

```
class Vehicle {
    void start() {
        System.out.println("Vehicle starts");
    }
}

class Car extends Vehicle {
    void drive() {
        System.out.println("Car drives");
    }
}

class SportsCar extends Car {
    void turbo() {
        System.out.println("Sports car turbo
mode ON");
    }
}

public class Main {
    public static void main(String[] args) {
        SportsCar sc = new SportsCar();
        sc.start();
        sc.drive();
        sc.turbo();
    }
}
```



Java

```
class Shape {
    void draw() {
        System.out.println("Drawing a shape");
    }
}

class Circle extends Shape {
    @Override
    void draw() {
        System.out.println("Drawing a circle");
    }
}

public class Main {
    public static void main(String[] args) {
        Shape s = new Circle();
        s.draw(); // Calls Circle's draw()
    }
}
```



Java

```
abstract class Animal {
    abstract void sound();

    void sleep() {
        System.out.println("Animal sleeps");
    }
}

class Cat extends Animal {
    @Override
    void sound() {
        System.out.println("Cat meows");
    }
}

public class Main {
    public static void main(String[] args) {
        Cat cat = new Cat();
        cat.sound();
        cat.sleep();
    }
}
```



Java

```
interface Vehicle {
    void start();
    void stop();
}

class Bike implements Vehicle {
    public void start() {
        System.out.println("Bike starts");
    }

    public void stop() {
        System.out.println("Bike stops");
    }
}

public class Main {
    public static void main(String[] args) {
        Bike bike = new Bike();
        bike.start();
        bike.stop();
    }
}
```



Java

```
public class PalindromeString {  
    public static void main(String[] args) {  
        String str = "madam";  
        String reversed = new  
StringBuilder(str).reverse().toString();  
  
        if (str.equals(reversed)) {  
            System.out.println(str + " is a  
palindrome");  
        } else {  
            System.out.println(str + " is not a  
palindrome");  
        }  
    }  
}
```



Java

```
public class PalindromeStringLoop {  
    public static void main(String[] args) {  
        String str = "level";  
        boolean isPalindrome = true;  
        int n = str.length();  
  
        for (int i = 0; i < n / 2; i++) {  
            if (str.charAt(i) != str.charAt(n -  
i - 1)) {  
                isPalindrome = false;  
                break;  
            }  
        }  
  
        System.out.println(str +  
(isPalindrome ? " is a palindrome" : " is not a  
palindrome"));  
    }  
}
```



Java

```
import java.util.Scanner;

public class VowelCheck {
    public static void main(String[] args) {
        char ch = 'a';
        if (ch == 'a' || ch == 'e' || ch == 'i'
|| ch == 'o' || ch == 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I'
|| ch == 'O' || ch == 'U') {
            System.out.println(ch + " is a
vowel");
        } else {
            System.out.println(ch + " is not a
vowel");
        }
    }
}
```



Java

```
public class VowelInString {  
    public static void main(String[] args) {  
        String str = "Hello World";  
        int count = 0;  
  
        for (char ch :  
str.toLowerCase().toCharArray()) {  
            if (ch == 'a' || ch == 'e' || ch ==  
'i' || ch == 'o' || ch == 'u') {  
                count++;  
            }  
        }  
  
        System.out.println("Number of vowels: "  
+ count);  
    }  
}
```



Java

```
public class EmployeeIDCheck {  
    public static void main(String[] args) {  
        String empId = "EMP12345";  
  
        if (empId.startsWith("EMP")) {  
            System.out.println(empId + " is a  
valid employee ID");  
        } else {  
            System.out.println(empId + " is  
invalid");  
        }  
    }  
}
```



Java

```
public class EmployeeIDRecovery {  
    public static void main(String[] args) {  
        String[] ids = {"EMP100", "EMP101",  
"EMP102"};  
        String searchId = "EMP101";  
        boolean found = false;  
  
        for (String id : ids) {  
            if (id.equals(searchId)) {  
                found = true;  
                break;  
            }  
        }  
  
        System.out.println(searchId + (found ?  
" found" : " not found"));  
    }  
}
```



Java

```
public class ReverseString {  
    public static void main(String[] args) {  
        String str = "Hello";  
        String reversed = new  
StringBuilder(str).reverse().toString();  
        System.out.println("Reversed String: "  
+ reversed);  
    }  
}
```



Java

```
public class ReverseStringLoop {  
    public static void main(String[] args) {  
        String str = "World";  
        String reversed = "";  
  
        for (int i = str.length() - 1; i >= 0;  
i--) {  
            reversed += str.charAt(i);  
        }  
  
        System.out.println("Reversed String: "  
+ reversed);  
    }  
}
```



Java

```
public class ArrayExample {  
    public static void main(String[] args) {  
        int[] numbers = {1, 2, 3, 4, 5};  
  
        for (int num : numbers) {  
            System.out.print(num + " ");  
        }  
    }  
}
```



Java

```
public class ArraySum {  
    public static void main(String[] args) {  
        int[] numbers = {10, 20, 30, 40};  
        int sum = 0;  
  
        for (int num : numbers) {  
            sum += num;  
        }  
  
        System.out.println("Sum: " + sum);  
    }  
}
```



Java

```
import java.util.Arrays;

public class SecondLargest {
    public static void main(String[] args) {
        int[] numbers = {10, 50, 20, 40, 30};
        Arrays.sort(numbers);
        System.out.println("Second Largest: " +
numbers[numbers.length - 2]);
    }
}
```



Java

```
public class SecondLargestNoSort {  
    public static void main(String[] args) {  
        int[] numbers = {10, 50, 20, 40, 30};  
        int largest = Integer.MIN_VALUE;  
        int secondLargest = Integer.MIN_VALUE;  
  
        for (int num : numbers) {  
            if (num > largest) {  
                secondLargest = largest;  
                largest = num;  
            } else if (num > secondLargest &&  
num != largest) {  
                secondLargest = num;  
            }  
        }  
  
        System.out.println("Second Largest: " +  
secondLargest);  
    }  
}
```